CS-550 PA3
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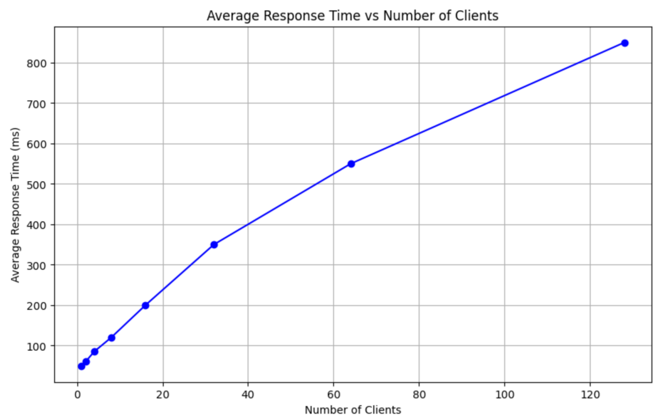
Performance Evaluation

Tree Topology Evaluations:

Experiment:

Executing 9 super-peers with each peer connected to 1-3 peers. Peers host files, which can be duplicated too across different peer nodes. The experiment is repeated with varying numbers of clients/peer nodes and the average response time is calculated for plotting as below:



Observation:

As the number of clients increases, the response time is expected to increase. At first, with a small number of clients, the system can handle incoming queries efficiently. But as the number of clients increases, the system may experience resource contention. Beyond a certain point, the increase in response time accelerates, indicating that the system is reaching its capacity. Eventually, the nodes at higher levels of the tree (closer to the root) may become bottlenecks, leading to increased response times. The actual shape of the curve will vary based on network load and resource limitations. Systems designed with scalability in mind may show a more gradual increase, while those with bottlenecks or less efficient resource utilization will show a steeper increase in response time. Response times might be more variable depending on the branch of the tree. Some branches may experience higher load and therefore higher response times if the tree is not balanced.

All-to-all Topology Evaluations:

Experiment:

Executing 9 super-peers with each peer connected to 1-3 peers in an all-to-all topology. Peers host files, which can be duplicated too across different peer nodes. The experiment is repeated with varying numbers of clients/peer nodes and the average response time is calculated for plotting as below:

<u>Observation:</u>

In an all-to-all topology, each node is connected to every other node in the network. With a few clients, the average response time might be low, as the network is underutilized, and each node can quickly respond to queries. The total number of connections increases rapidly with the increase in the number of nodes, following the formula N(N - 1) / 2, where N represents the number of nodes in the network. This rapid increase in connections results in higher overall network traffic and can quickly saturate the network bandwidth compared to a tree topology. As more clients send queries simultaneously, the number of concurrent operations that each node must handle increases dramatically. This can overwhelm the processing capabilities of the nodes and slow down response times due to processing queue buildup. Many other factors like resource contention, network overhead, or propagation delay might impact the response time for all-to-all topology.

Tree vs All-to-all topology

To summarize, in an all-to-all topology, the average response time is expected to increase at a faster rate as the number of clients increases. This is because the number of connections grows quadratically. On the other hand, in a tree topology, the increase in response time is expected to be more gradual since the connections are formed in a more controlled manner. However, there may still be bottlenecks at higher levels of the tree hierarchy.